

IIE6120 Linear Programming
2025-2 Homework 02
Team B (이정민, 장서현)

Homework 2

Q1. 2.16

Let $B \in \mathbb{R}^{n \times n}$ and $\tilde{B} \in \mathbb{R}^{n \times n}$ where the i th row of B is multiplied by scalar $\lambda \neq 0$.

$\tilde{B} = E_i(\lambda) B$ where $E_i(\lambda) = \text{diag}(1, \dots, 1, \lambda, 1, \dots, 1)$

$\tilde{B}^{-1} = [E_i(\lambda) B]^{-1} = B^{-1} E_i(\lambda)^{-1}$

Since $E_i(\lambda)$ is diagonal, $E_i(\lambda)^{-1} = \text{diag}(1, \dots, 1, \frac{1}{\lambda}, 1, \dots, 1)$

Thus $\tilde{B}^{-1} = B^{-1} \text{diag}(1, \dots, 1, \frac{1}{\lambda}, 1, \dots, 1)$

$\therefore$ Only the i th column of B^{-1} is multiplied by $1/\lambda$. All the other columns remain unchanged.

Q2. 2.17

Let $B \in \mathbb{R}^{n \times n}$ and $\tilde{B} \in \mathbb{R}^{n \times n}$ where the i th col. of B is multiplied by scalar $\lambda \neq 0$.

$\tilde{B} = B E_i(\lambda)$ where $E_i(\lambda) = \text{diag}(1, \dots, 1, \lambda, 1, \dots, 1)$

$\tilde{B}^{-1} = [B E_i(\lambda)]^{-1} = E_i(\lambda)^{-1} B^{-1}$

Since $E_i(\lambda)$ is diagonal, $E_i(\lambda)^{-1} = \text{diag}(1, \dots, 1, \frac{1}{\lambda}, 1, \dots, 1)$

Thus $\tilde{B}^{-1} = \text{diag}(1, \dots, 1, \frac{1}{\lambda}, 1, \dots, 1) B^{-1}$

$\therefore$ Only the i th row of B^{-1} is multiplied by $1/\lambda$. All the other rows remain unchanged.

Q3. 2.18

$$(i) \begin{bmatrix} 1 & 0 & 1 & -1 \\ 2 & 3 & 4 & 1 \\ 1 & 0 & 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 3 & 2 & 3 \\ 0 & 0 & 4 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 2/3 & 1 \\ 0 & 0 & 1 & -1/2 \end{bmatrix}$$

$\therefore \text{rank}(A) = 3$

$$(ii) \begin{bmatrix} -2 & 1 & -3 \\ 2 & 4 & -2 \\ 4 & 3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & -3 \\ 0 & 5 & -5 \\ 4 & 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 1 & -3 \\ 0 & 1 & -1 \\ 0 & 5 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 1 & -3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1/2 & 3/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore \text{rank}(A) = 2$

Q4. 2.21

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ -1 & 1 & 0 & 0 & 1 \\ 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 2 \\ -3 \\ -2 \end{bmatrix}$$

Basis $B \in \mathbb{R}^{4 \times 4}$.
 a_3 must be included
 and since a_1, a_2, a_4 are linearly dependent,
 $(\because a_4 = a_1 + a_2)$
 a_5 must also be included.

① exclude a_1

$$B = [a_2 \ a_3 \ a_4 \ a_5] \sim [a_3 \ a_4 \ a_2 \ a_5]$$

$$= \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & -1 & -1 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$\therefore$ triangular

② exclude a_2

$$B = [a_1 \ a_3 \ a_4 \ a_5] \sim [a_4 \ a_3 \ a_1 \ a_5]$$

$$= \begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$\therefore$ triangular

③ exclude a_4

$$B = [a_1 \ a_2 \ a_3 \ a_5] \sim [a_2 \ a_3 \ a_1 \ a_5]$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$\therefore$ triangular

Q5. 2.24

$$\left[\begin{array}{cccc|c} -1 & 2 & 1 & 3 & -2 & 4 \\ 1 & -2 & 0 & 2 & 1 & 2 \end{array} \right] = \left[\begin{array}{cccc|c} -1 & 2 & 1 & 3 & -2 & 4 \\ 0 & 0 & 1 & 5 & -1 & 6 \end{array} \right]$$

possible ones (rank=2):

$$(x_1, x_3) \begin{cases} -x_1 + x_3 = 4 \\ x_3 = 6 \end{cases} \quad (2, 0, 6, 0, 0)$$

$$(x_1, x_4) \begin{cases} -x_1 + 3x_4 = 4 \\ 5x_4 = 6 \end{cases} \quad (-\frac{2}{5}, 0, 0, \frac{6}{5}, 0)$$

$$(x_1, x_5) \begin{cases} -x_1 - 2x_5 = 4 \\ -x_5 = 6 \end{cases} \quad (8, 0, 0, 0, -6)$$

$$(x_2, x_3) \begin{cases} 2x_2 + x_3 = 4 \\ x_3 = 6 \end{cases} \quad (0, -1, 6, 0, 0)$$

$$(x_2, x_4) \begin{cases} 2x_2 + 3x_4 = 4 \\ 5x_4 = 6 \end{cases} \quad (0, \frac{1}{5}, 0, \frac{6}{5}, 0)$$

$$(x_2, x_5) \begin{cases} 2x_2 - 2x_5 = 4 \\ -x_5 = 6 \end{cases} \quad (0, -4, 0, 0, -6)$$

$$(x_3, x_4) \begin{cases} x_3 + 3x_4 = 4 \\ x_3 + 5x_4 = 6 \end{cases} \quad (0, 0, 1, 1, 0)$$

$$(x_3, x_5) \begin{cases} x_3 - 2x_5 = 4 \\ x_3 - x_5 = 6 \end{cases} \quad (0, 0, 8, 0, 2)$$

$$(x_4, x_5) \begin{cases} 3x_4 - 2x_5 = 4 \\ 5x_4 - x_5 = 6 \end{cases} \quad (0, 0, 0, \frac{8}{11}, -\frac{2}{11})$$

Q6. 2.25

$$\left[\begin{array}{cccc|c} 1 & 3 & 1 & -1 & 1 \\ 0 & 5 & -6 & 1 & 0 \\ 1 & -2 & 4 & 0 & 2 \end{array} \right] = \left[\begin{array}{cccc|c} 1 & 3 & 1 & -1 & 1 \\ 0 & 5 & -6 & 1 & 0 \\ 0 & -5 & 3 & 1 & 1 \end{array} \right]$$

$$= \left[\begin{array}{cccc|c} 1 & 3 & 1 & -1 & 1 \\ 0 & 5 & -6 & 1 & 0 \\ 0 & 0 & -3 & 2 & 1 \end{array} \right]$$

it has infinitely many sol since $n=4$ rank=3.

Q7. 2.28

(a)  $\therefore$ not convex.

(b) let there are two points (a, b, c) and (d, e, f) which satisfies

$$\begin{cases} a+2b \leq 1 \\ a-2c \leq 2 \end{cases} \quad \text{and} \quad \begin{cases} d+2e \leq 1 \\ d-2f \leq 2 \end{cases}$$

let there is a point $(\lambda a + (1-\lambda)d, \lambda b + (1-\lambda)e, \lambda c + (1-\lambda)f), 0 \leq \lambda \leq 1$

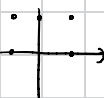
which are located between two points aforementioned.

we know $\begin{cases} \lambda(a+2b) \leq \lambda \\ \lambda(a-2c) \leq 2\lambda \end{cases}$ and $\begin{cases} (1-\lambda)(d+2e) \leq (1-\lambda) \\ (1-\lambda)(d-2f) \leq 2(1-\lambda) \end{cases}$

and $\lambda a + (1-\lambda)d + 2\lambda b + 2(1-\lambda)e \leq 1$
 $\lambda a + (1-\lambda)d - 2\lambda c - 2(1-\lambda)f \leq 2$

$$\Rightarrow \begin{cases} \lambda a + (1-\lambda)d + 2\lambda b + 2(1-\lambda)e \leq 1 \\ \lambda a + (1-\lambda)d - 2\lambda c - 2(1-\lambda)f \leq 2 \end{cases}$$

$\therefore$ convex

(c)  $A = (1, 3) \quad B = (-1, 3)$
 mid-point = $(0, 3)$

$A, B \in C$, but $(0, 3) \Rightarrow 3-0=3 \neq 0$

$\therefore$ NOT convex.

(d) let there are two points (a,b,c) & (d,e,f) which satisfies

$$\begin{cases} b \geq a^2 \\ a+2b+c \leq 4 \end{cases} \quad \begin{cases} e \geq d^2 \\ d+2e+f \leq 4 \end{cases}$$

and $\exists$ a point between two which is

$$(\lambda a + (1-\lambda)d, \lambda b + (1-\lambda)e, \lambda c + (1-\lambda)f), \quad 0 \leq \lambda \leq 1$$

we aware $\begin{cases} \lambda b \geq \lambda a^2 \\ \lambda(a+2b+c) \leq \lambda 4 \end{cases} \quad \begin{cases} (1-\lambda)e \geq (1-\lambda)d^2 \\ (1-\lambda)(d+2e+f) \leq (1-\lambda)4 \end{cases}$

① $\lambda a + (1-\lambda)d + 2\lambda b + 2(1-\lambda)e + \lambda c + (1-\lambda)f \leq 4$

② $\lambda b + (1-\lambda)e \geq \lambda a^2 + (1-\lambda)d^2 \Leftrightarrow (\lambda a + (1-\lambda)d)^2$

$$\begin{matrix} \text{--- (A)} & \text{--- (B)} \\ \lambda a^2 + 2\lambda(1-\lambda)ad + (1-\lambda)d^2 \end{matrix}$$

$$\begin{aligned} [A-B] &= a^2(\lambda - \lambda^2) + d^2(1-\lambda - (1-\lambda)^2) - 2\lambda(1-\lambda)ad \\ &= \lambda(1-\lambda)a^2 + \lambda(1-\lambda)d^2 - \lambda(1-\lambda)2ad \\ &= \lambda(1-\lambda)(a^2 + d^2 - 2ad) \\ &= \lambda(1-\lambda)(a-d)^2 \geq 0 \end{aligned}$$

$$\therefore \lambda a^2 + (1-\lambda)d^2 \geq (\lambda a + (1-\lambda)d)^2$$

$$\therefore \lambda b + (1-\lambda)e \geq (\lambda a + (1-\lambda)d)^2$$

$\Rightarrow$ convex.

(e) assume there are two points in the set, $(3,a)$ $(3,b)$

and $\exists$ a point between two, $(3, \lambda a + (1-\lambda)b)$ $0 \leq \lambda \leq 1$.

it is known that $\begin{cases} |a| \leq 4 \\ |b| \leq 4 \end{cases} \Rightarrow \begin{cases} |\lambda a| \leq \lambda 4 \\ |\lambda b| \leq \lambda 4 \end{cases}$

then $|\lambda a| + |\lambda b| \leq \lambda 4$

$$|\lambda a + (1-\lambda)b| \leq 4$$

$\therefore$ convex.

(f) $(0,1,1)$ and $(0,-1,1)$ satisfy the set.

$$\begin{cases} |x| = 1 \\ 0 \leq z \end{cases} \quad \begin{cases} |x| = 1 \\ 0 \leq z \end{cases}$$

but the mid point $(0,0,1)$ does NOT. $\begin{cases} |x| \neq 1 \\ 0 \leq z \end{cases}$

$\therefore$ not convex.

Q8. 2.39

By definition, a CONVEX CONE C is a convex set with the additional property that $\lambda x \in C$ for each $x \in C$ and for each $\lambda \geq 0$.

Therefore, C always contain the origin by letting $\lambda = 0$, and the ray $\{\lambda x : \lambda \geq 0\}$ belongs to C .

Actually the ray in C is expressed in the strict convex comb of two distinct points which contradicts to the def of extreme points

Another example is this:

Assume there exists an extreme point $x \in C$ with $x \neq 0$

Because C is a convex cone, ① $x=0 \Rightarrow 0 \in C$
 $\alpha x \in C$ ② $\alpha = 2 \Rightarrow 2x \in C$

Taking $\lambda = \frac{1}{2}$, x can be written in convex combination.

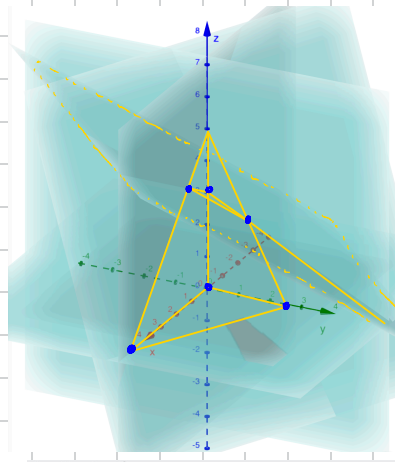
$$x = \frac{1}{2}(0) + \frac{1}{2}(2x)$$

For x to become an extreme point, $x=0=2x$ must hold.

thus $x=0$, which contradicts our assumption that $x \neq 0$.

$\therefore$ origin is the ONLY extreme point of convex cone.

Q9. 2.40



let $x_1 + 2x_2 + x_3 = 5 \rightarrow$ (A)

$-x_1 + x_2 + 2x_3 = 6 \rightarrow$ (B)

$x_1 = 0 \rightarrow$ (C)

$x_2 = 0 \rightarrow$ (D)

$x_3 = 0 \rightarrow$ (E)

1) A, B, C

$x_1 = 0, x_2 = 4/3, x_3 = 7/3$

2) A, B, D

$x_1 = 4/3, x_2 = 0, x_3 = 11/3$

3) A, B, E

$x_1 = -7/3, x_2 = 11/3, x_3 = 0 \rightarrow$ (C violation)

4) A, C, D

$x_1 = 0, x_2 = 0, x_3 = 5 \rightarrow$ (B violation)

5) A, C, E

$x_1 = 0, x_2 = 5/2, x_3 = 0$

6) A, D, E

$x_1 = 5, x_2 = 0, x_3 = 0$

7) B, C, D

$x_1 = 0, x_2 = 0, x_3 = 3$

8) B, C, E

$x_1 = 0, x_2 = 6, x_3 = 0 \rightarrow$ (A violation)

9) B, D, E

$x_1 = -6, x_2 = 0, x_3 = 0 \rightarrow$ (C violation)

10) C, D, E

$x_1 = 0, x_2 = 0, x_3 = 0$

Q10. 2.41

The following set has directions as x_3 can be increased unboundedly within the set's feasible region.

The recession cone of X : $D = \{d : -d_1 + 2d_2 = 0, 2d_1 - 2d_2 - d_3 \leq 0, d_1, d_2, d_3 \geq 0\}$

$$\downarrow d_1 = 2d_2 \rightarrow 2d_2 - d_3 \leq 0$$

$$= \{d : d_1 = 2d_2, d_3 \geq 2d_2, d_1, d_2, d_3 \geq 0\}$$

Using the normalization constraint $D' = \{d \in D, d_1 + d_2 + d_3 = 1\}$

$$\hookrightarrow 3d_2 + d_3 = 1$$

① $3d_2 + d_3 = 1, d_1 = 2d_2, d_1 = 0 \rightarrow (d_1, d_2, d_3) = (0, 0, 1)$

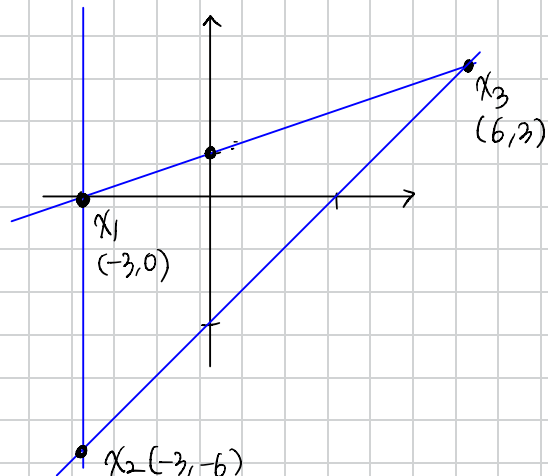
② $3d_2 + d_3 = 1, d_1 = 2d_2, d_2 = 0 \rightarrow \quad \quad \quad = (0, 0, 1)$

③ $3d_2 + d_3 = 1, d_1 = 2d_2, d_3 = 0 \rightarrow \quad \quad \quad = (\frac{2}{3}, \frac{1}{3}, 0) \rightarrow$ violates $2d_2 - d_3 \leq 0$.

④ $3d_2 + d_3 = 1, d_1 = 2d_2, 2d_2 - d_3 = 0 \rightarrow \quad \quad \quad = (\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$

$\therefore$ E.D. : $(0, 0, 1), (\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$

Q11. 2.48



$\therefore x = (0,1)$ can be expressed as

$$\frac{2}{9}x_1 + 0x_2 + \frac{1}{9}x_3$$

Q12.

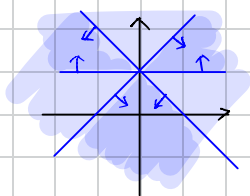
(a) True.

Consider $X = \{x : Ax \leq b, x \geq 0\}$ where $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

For x , we need $x_1 \leq 0, x_1 \geq 1 \rightarrow$ empty set $X = \emptyset$

For D , $D = \{d : Ad \leq 0, d \geq 0, Id = 1\}$ we can take $d = (0, 1)$
 $D \neq \emptyset$

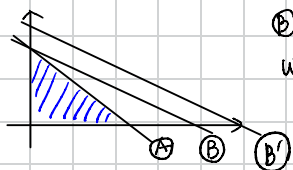
(b) suppose there exists such condition.



then the degeneracy occurred
but NOT redundant.

so degeneracy NOT always lead to
redundancy.

(c) False



ⓑ constraint is redundant.
with degenerate point

but as ⓑ shifts to ⓑ'
the redundancy still exists,
but degeneracy isn't.

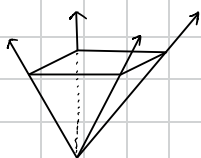
(d) False

$x = (x_1, x_2, x_3)$

where $x_1 \geq 0$, and $x_2 \leq 0$

it is not empty without extreme point.

(e) False



example of a 3dim unbounded.
polyhedral set having + extreme direct.

(f) $n-r$

when the number of unknowns are n and

$Ax=b$ enforces r var. the maximal dim is $n-r$.